INCH-POUND
MIL-C-62/7C
12 February 2003
SUPERSEDING
MIL-C-62/7B
19 December 1968

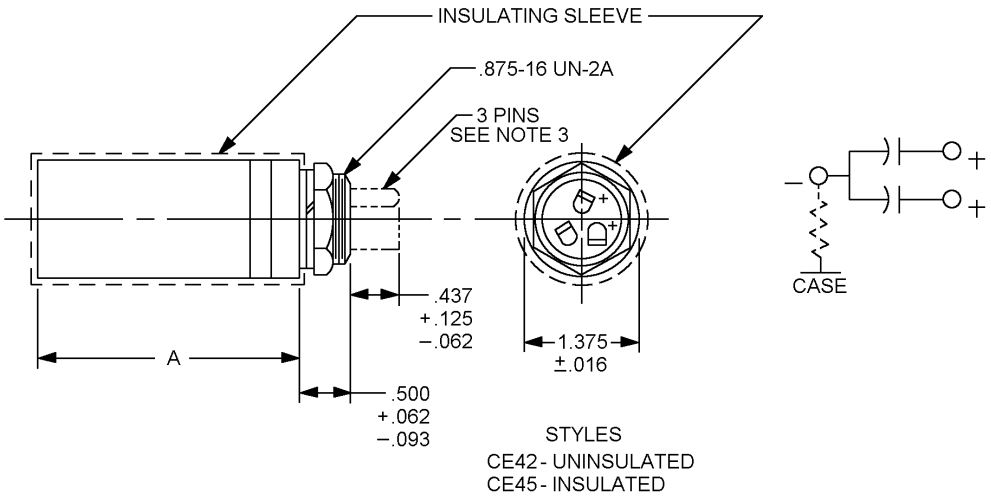
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC (DC, ALUMINUM, DRY ELECTROLYTE, POLARIZED) STYLES CE42 AND CE45

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-C-62.

Inactive for design after 27 March 1979



Inches	mm
.016	.41
.062	1.57
.093	2.36
.125	3.18
.437	11.10
.500	12.70
1.375	34.93

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Shape and position of lugs optional.
4. Mounting nut and lockwasher shall be supplied with each capacitor (Family 3).
5. There is an indeterminate resistance between the metal case and the negative terminal.

FIGURE 1. Dimensions and configuration.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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TABLE I. Styles CE42 and CE45 capacitors.

PIN 1/	Capacitance 4/	DC rated voltage per section	DC leakage per section	Maximum impedance per section	Dissipation factor per section	Case dimensions 2/ A 3/ ± .062 (1.57)
	uF	volts	uA	ohms	percent	inches
CE4-C330J	33-33	150	106	266	12	2.250 (57.15)
CE4-C680J	68-68	150	151	129	12	2.250 (57.15)
CE4-C750J	75-75	150	160	117	12	2.250 (57.15)
CE4-C151J	150-150	150	225	59	12	3.000 (76.20)
CE4-C221J	220-220	150	273	40	12	3.750 (95.15)
CE4-C620K	62-62	200	166	142	12	2.250 (57.15)
CE4-C680K	68-68	200	175	129	12	2.500 (63.50)
CE4-C750K	75-75	200	183	117	12	2.500 (63.50)
CE4-C101K	100-100	200	211	88	12	2.750 (69.85)
CE4-C151K	150-150	200	260	49	12	3.250 (82.55)
CE4-C300M	30-30	250	129	293	12	2.250 (57.15)
CE4-C470M	47-47	250	166	187	12	2.250 (57.15)
CE4-C560M	56-56	250	178	157	12	2.500 (63.50)
CE4-C680M	68-68	250	195	129	12	2.750 (69.85)
CE4-C750M	75-75	250	206	117	12	3.000 (76.20)
CE4-C101M	100-100	250	237	88	12	3.250 (82.55)
CE4-C360N	36-36	300	156	244	12	2.250 (57.15)
CE4-C680N	68-68	300	214	129	12	3.250 (82.55)
CE4-C101N	100-100	300	260	88	12	4.250 (107.95)

See footnotes at end of table.

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TABLE I. Styles CE42 and CE45 capacitors - Continued.

PIN 1/	Capacitance 4/	DC rated voltage per section	DC leakage per section	Maximum impedance per section	Dissipation factor per section	Case dimensions 2/ A 3/ ± .062 (1.57)
	uF	volts	uA	ohms	percent	Inches
CE4-C100P	10-10	350	88	880	12	2.250 (57.15)
CE4-C200P	20-20	350	125	440	12	2.250 (57.15)
CE4-C300P	30-30	350	154	293	12	2.250 (57.15)
CE4-C470P	47-47	350	192	187	12	2.750 (69.85)
CE4-C620P	62-62	350	222	142	12	3.250 (82.55)
CE4-C680P	68-68	350	232	129	12	3.750 (95.25)
CE4-C100Q	10-10	400	94	1480	17	2.250 (57.15)
CE4-C150Q	15-15	400	116	986	17	2.250 (57.15)
CE4-C220Q	22-22	400	141	670	17	2.250 (57.15)
CE4-C330Q	33-33	400	172	450	17	3.000 (76.20)
CE4-C470Q	47-47	400	206	314	17	3.500 (88.90)
CE4-C100R	10-10	100	100	2200	17	2.250 (57.15)
CE4-C150R	15-15	450	123	1465	17	2.250 (57.15)
CE4-C220R	22-22	450	149	1000	17	2.750 (69.85)
CE4-C300R	30-30	450	174	733	17	3.500 (88.90)
CE4-C430R	43-43	450	208	512	17	3.750 (95.25)

1/ Complete type designation shall include an additional digit to complete style.

2/ Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

3/ The dimensions shown are for style CE42; for style CE45; add .062 (1.57) ± .062 (1.57) to the nominal dimension.

4/ There are two identical separate capacitors contained in each capacitor case.

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REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

Case type: Tubular, uninsulated (CE42) and insulated (CE45).

Capacitance value: See table I.

Capacitance tolerance:

At 200 volts, dc rated, and above: -10, +50 percent.

At 150 volts, dc rated, and less: -10, +75 percent

Operating temperature range: -40ºC to +85ºC.

Solderability: Method 208 of MIL-STD-202.

DC leakage: See table I.

Capacitance: Method 305 of MIL-STD-202.

Dissipation factor: See table I.

Terminal strength: Method 211 of MIL-STD-202, test conditions A, B, and E.

Shock, medium impact: Method 205, test condition C, MIL-STD-202.

Vibration, high frequency: Method 204, test condition B, MIL-STD-202.

Salt atmosphere (corrosion): Method 101, test condition B, MIL-STD-202.

Temperature cycling and immersion:

Temperature cycling: Method 102, test condition D, MIL-STD-202.

Immersion: Method 104, test condition C, MIL-STD-202.

DWV: Insulating sleeves - Method 301 of MIL-STD-202. 4,000 volts dc applied.

DCL: Not more than 150 percent of initial measurement.

Capacitance change: 5 percent.

DF: See table I.

Corrosion: Not more than 10 percent.

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Moisture resistance: Method 106 of MIL-STD-202.

Insulation resistance (insulating sleeve): Not less than 100 megohms.

DWV: Insulating sleeves - Method 301 of MIL-STD-202. 4,000 volts dc applied.

DCL: Not more than 150 percent of initial requirement.

DC: See table I.

Corrosion: Not more than 10 percent.

Stability at low and high temperature: In accordance with MIL-C-62.

Step 1 (at + 25ºC).

DCL: See table I.

Capacitance: See table I.

DF: See table I.

Step 2 (at -40ºC): Impedance - See table I.

Step 3 (at +25ºC):

DCL: See table I.

Capacitance: ± 5 percent of initial value.

DF: See table I.

Step 4 (at +85ºC):

DCL: Not more than 10 times initial value, or meet initial 25ºC requirement.

Capacitance - ± 20 percent of initial value.

DF: See table I.

Step 5 (at +25ºC):

DCL: See table I.

Capacitance - ± 5 percent of initial value.

DF: See table I.

Surge voltage: In accordance with MIL-C-62.

DCL: See table I.

Capacitance change: 6 percent.

DF: See table I.

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Vent: Not applicable.

Life: Method 108, test condition F, MIL-STD-202.

DCL: See table I.

Capacitance change: 100 volts, dc rated, and less: -20, + 15 percent.

DC rated voltage above 100 volts: ±15 percent.

DF: Meet the initial requirement or not exceed 150 percent of the initial measurement.

Barometric pressure: Method 105 of MIL-STD-202.

Shelf life: In accordance with MIL-C-62.

Build factor: Not more than 4.0.

Shelf factor: Not more than 3.0.

DCL: Not more than 150 percent of initial value.

Capacitance change: 6 percent.

DF: Not more than 175 percent of initial requirement.

Application data:

All capacitors shall withstand an rms ripple current at 120 Hz and 85ºC as determined from the formula:

$$I_r = k\sqrt{C}$$

Where:

I_r = rms ripple current in milliamps.
C = normal capacitance in microfarads per section.
k = 50 for units rated at 150 and 200 volts dc.
= 35 for units rated at 25o to 350 volts dc, incl.
= 30 for units rated at 400 to 450 volts dc.

QUALIFICATION ASSURANCE PROVISIONS:

Qualification inspection: Not applicable.

Retention of qualification: Not applicable.

Conformance inspection: Group A inspection only.

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TABLE II. Supersession data.

Superseded insulated PIN	Superseded uninsulated PIN	Recommended replacement style CE42 or CE45	Superseded insulated PIN	Superseded uninsulated PIN	Recommended replacement style CE42 or CE45
001	078	CE4-C100P	027	104	CE4-C101M
002	079	CE4-C300M	028	105	CE4-C100P
003	080	CE4-C330J	029	106	CE4-C300P
004	081	CE4-C470M	030	107	CE4-C360N
005	082	CE4-C680J	031	108	CE4-C470P
006	083	CE4-C750J	032	109	CE4-C620P
007	084	CE4-C101K	033	110	CE4-C680N
008	085	CE4-C151J	034	111	CE4-C101N
009	086	CE4-C221J	035	112	CE4-C100P
010	087	CE4-C100P	036	113	CE4-C200P
011	088	CE4-C300M	037	114	CE4-C300P
012	089	CE4-C360N	038	115	CE4-C470P
013	090	CE4-C470M	039	116	CE4-C470P
014	091	CE4-C470M	040	117	CE4-C620P
015	092	CE4-C620K	041	118	CE4-C680P
016	093	CE4-C680K	042	119	CE4-C100Q
017	094	CE4-C750K	043	120	CE4-C150Q
018	095	CE4-C101K	044	121	CE4-C220Q
019	096	CE4-C151K	045	122	CE4-C330Q
020	097	CE4-C100P	046	123	CE4-C470Q
021	098	CE4-C300M	047	124	CE4-C100R
022	099	CE4-C300M	048	125	CE4-C150R
023	100	CE4-C470M	049	126	CE4-C220R
024	101	CE4-C560M	050	127	CE4-C300R
025	102	CE4-C680M	051	128	CE4-C430R
026	103	CE4-C750M			

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - AT, AV, CR4, MI
Navy - AS, MC, OS
Air Force - 19, 99

Preparing activity:
DLA - CC

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